

Madhav Malhotra

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Skills

Software

Python (**Langchain**, **FastAPI**, **Neo4j**, PyTorch, Numpy, Pandas, Celery), Typescript (**Next**, **trPC**, **Livekit**, Prisma), C/C++ (multi-threading, OOP), SQL (**PostgreSQL**, Supabase), Linux (Bash, VMs, Make), **Docker**, Kubernetes, **Azure** (ACR, Container Apps)

Hardware

VLSI (**Verilog/System Verilog**, cocotb, sby, openlane), MCU Platforms (**STM32-F401**, Arduino Uno, Raspberry Pico/Pi), ARM ISA/v7 assembly language, Lab experience (oscilloscopes, function generators, soldering, etc.)

Soft Skills

Public speaking, attention to detail, **management**, bias to action, cold-emailing, UX design. Languages: English (native), Hindi (native), French (DELF-B2)

Experience

Jurin.ai

ML Engineering Intern

(2025.12 - 2026.04)

Implemented **knowledge graph ingestion with neo4j**, **reducing latency 5x with multithreading and batching**. Created automated regression evaluations to discover production bug reducing retrieval accuracy by 70%. Implemented **realtime transcription and call transfer with Livekit voice agents and WebRTC** to enable live customer support monitoring.

Boutros Lab

RL Research Assistant

(2025.09 - 2025.12)

Implementing contextualised bandit RL agent in C++ to reduce place and route runtime in the Verilog to Routing (VTR) FPGA CAD suite. Applied **caching and quantisation to reduce runtime by 40%** without wirelength increase.

Pellizzoni Lab

RTL Research Assistant

(2025.05 - 2025.08)

Created a coherent RISC-V instruction cache for the Berkeley Out of Order Machine with Chisel and Chipyard. Researched the TileLink-C protocol to create specifications and state transition tables for the design.

Analyst3.ai

ML Intern

(2025.01 - 2025.04)

Ran prompt engineering and model evaluations in Promptlayer to lower latency by 60% for a financial research agent. **Enabled model completion streaming for 22x lower time to first load** with the Azure OpenAI service and a Next.js frontend. Migrated FastAPI/Celery/Supabase backend to Azure Container Apps with Github CI/CD actions and Bash scripting.

Health Infobase

ML Developer

(2024.05 - 2024.09)

Used Langchain and Faiss to create a chatbot to answer epidemiology questions. **Reduced token costs by 30% with RAG and CoT prompting**. Also surveyed benchmarks/pricing to select embedding models, vectorstore providers, etc. Deployed REST API to access the chatbot using scalable docker images on the AWS Lightsail Container Registry.

Projects

WAT.ai IoT Security

Led design team to **publish a conference paper (bit.ly/CUCAI) on IoT anomaly detection with genetic algorithms**. Used Numpy and sklearn to memoize and vectorise the DBSCAN algorithm, achieving 350x speedup. Used Numpy, PyTorch, & Pandas to apply a genetic optimiser to compress training data by 95% (bit.ly/wolfopt).

WAT.ai Policy Team

Led design team to **publish a conference paper (bit.ly/poliLLM) on LLM agents for legislative policy analysis**. Ran ablation tests to compare agentic architectures with reflective memories and roles like courtroom lawyers.

Grade 8 Math to AI

Made a **Youtube course teaching AI starting from grade 8 math**. Taught graphing, calculus, matrix operations, regression, gradient descent, neural networks, CNNs, Python/PyTorch/Pandas, and an **intro to Deep RL**.

Education

University of Waterloo

BASc, Computer Engineering

(2022.09 - 2027.06)

Extracurriculars: **VP, WAT.ai Design Team**. Capstone: building a Verilog Out-of-order processor for a game on computer architecture. Courses: ECE 327 (RTL), ECE 222/320 (Computer Architecture), ECE 351 (Compilers), ECE 252/350 (OS/kernels).

DeepLearning.ai

Took various courses on deep learning theory, from rudimentary regression to modern transformers. Projects included numpy neural networks, **CNN fine-tuning with PyTorch**, and **PEFT adapters for Google's flan-t5**.